

Randomized and low-rank approximation

HSS approximation from a number of matvecs independent of depth

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: Challenging because reusing sketches introduces dependencies across the hierarchy; community impact is reducing operator access in black-box HSS compression.

Area: randomized matrix compression

Context and notation

Use exact real arithmetic, with comparisons, standard Gaussian sampling, and exact SVDs available as primitives; charge an SVD of an $a \times b$ matrix $O(ab \min(a, b))$ operations. This states the idealized arithmetic model used here, rather than a finite precision or bit complexity claim. A matrix–vector query returns either Av or $A^\top v$ for one chosen real vector v ; both types count toward the total. Randomized guarantees are for every fixed input, with probability or expectation over the algorithm’s randomness.

For $N = 2^{L+1}k$, $k, L \geq 1$, form the perfect binary tree that successively halves the ordered index set $[N] = \{1, \dots, N\}$ until every leaf contains $2k$ indices. Define $\mathcal{S}_{L,k}$ to contain precisely the real $N \times N$ matrices B such that, for every nonroot node I of this tree,

$$\text{rank } B[I, [N] \setminus I] \leq k, \quad \text{rank } B[[N] \setminus I, I] \leq k.$$

This is the HSS class with rank at most k , including its constraints across levels. There is no restriction inside a leaf diagonal block. Set

$$E_{L,k}(A) = \min_{B \in \mathcal{S}_{L,k}} \|A - B\|_F^2.$$

The minimum exists. The rank definition follows Stefano Masei, Leonardo Robol, and Daniel Kressner, *hm-toolbox: MATLAB Software for HODLR and HSS Matrices*, SIAM Journal on Scientific Computing **42** (2020), C43–C68, §2.2, Definition 3. Equivalence with the telescoping representation and existence of a minimizer are recorded in Amsel et al., cited below, Definition 3 and Appendix D.

Problem statement

Do absolute constants $C_q, C_e, c > 0$ and a uniform randomized algorithm exist that, for every $k, L \geq 1$ and every $A \in \mathbb{R}^{N \times N}$, $N = 2^{L+1}k$, accesses A solely through at most $C_q k$ matrix–vector queries and returns $B \in \mathcal{S}_{L,k}$ satisfying

$$\mathbb{E} \|A - B\|_F^2 \leq C_e L E_{L,k}(A),$$

with at most $O(N^c)$ additional arithmetic operations? Queries may be adaptive. The budget counts both transpose and forward products and is a bound on every execution.

References

Amsel et al., *Quasi-optimal Hierarchically Semi-separable Matrix Approximation*, §1.2, Theorem 13, §§4.4–5. James Levitt and Per-Gunnar Martinsson, *Linear-Complexity Black-Box Randomized Compression of Rank-Structured Matrices*, SIAM Journal on Scientific Computing **46** (2024), A1747–A1763, Algorithm 4.1 and Remark 4.3.

Status evidence

The proved guarantee uses $O(kL)$ queries. Reusing sketches reduces the query count, but its approximation guarantee remains unproved. Searches for “HSS $O(k)$ approximation 2026 matvec” and “HSS approximation logarithmic 2026” found no resolution. Levitt–Martinsson’s latest arXiv version is v3, submitted June 21, 2024. Christopher Musco’s [February 2026 ICERM slides](#), numbered slide 14, still list $O(k \log N)$ queries for HSS approximation.

[RE-01](#) permits full access and asks for a constant error factor. RE-02 retains the known dependence of the error factor on depth and asks for fewer queries. Neither target directly implies the other.

Audit — 2026-09-10

Rechecked [Amsel et al., Theorem 13 and §4.4](#). The proved approximation uses depth-dependent queries; the reused-sketch algorithm lacks the requested guarantee. HSS matvec and later-publication searches found no resolution with a query count independent of depth.